

Review Session 3

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1 Linear Transformation

Theorem 1. Let (X_t) be a stationary process. Let (Y_t) be a filtered process of (X_t) given by

$$Y_t = \sum_{k=-\infty}^{\infty} \alpha_k X_{t-k} = \alpha(L)X_t$$

where $\alpha(L) = \sum_{k=-\infty}^{\infty} \alpha_k L^k$ with $\sum_{k=-\infty}^{\infty} |\alpha_k| < \infty$. Let the spectral representation of X_t and Y_t be given by

$$\begin{aligned} X_t &= \int_{-\pi}^{\pi} e^{-it\lambda} dZ_X(\lambda) \\ Y_t &= \int_{-\pi}^{\pi} e^{-it\lambda} dZ_Y(\lambda) \end{aligned}$$

Then, we have

$$Z_Y(\Lambda) = \int_{\Lambda} e^{-it\lambda} \alpha(e^{i\lambda}) dZ_X(\lambda)$$

for all $\Lambda \in \mathcal{B}$. Also, we have

$$f_Y(\lambda) = |\alpha(e^{i\lambda})|^2 f_X(\lambda)$$

if $F_Y(\lambda)$ and $F_X(\lambda)$ are absolutely continuous w.r.t. Lebesgue measure.

Proof. Let $\alpha_n(L) = \sum_{k=-n}^n \alpha_k L^k$. Consider the truncated version of the filtered process (Y_t) given by $\alpha_n(L)X_t = \sum_{k=-n}^n \alpha_k X_{t-k}$. Using spectral representations of (X_t) , we have

$$\begin{aligned} \sum_{k=-n}^n \alpha_k X_{t-k} &= \sum_{k=-n}^n \alpha_k \int_{-\pi}^{\pi} e^{-i(t-k)\lambda} dZ_X(\lambda) \\ &= \int_{-\pi}^{\pi} e^{-it\lambda} \sum_{k=-n}^n e^{ik\lambda} \alpha_k dZ_X(\lambda) \end{aligned}$$

Notice that $\sum_{k=-n}^n e^{ik\lambda} \alpha_k$ is dominated by $\sum_{k=-\infty}^{\infty} |\alpha_k|$ for all $\lambda \in [-\pi, \pi]$ and for all $n \in \mathbb{N}$. Thus, using dominated convergence theorem, we have

$$\begin{aligned} Y_t &= \sum_{k=-\infty}^{\infty} \alpha_k X_{t-k} = \int_{-\pi}^{\pi} e^{-it\lambda} \sum_{k=-\infty}^{\infty} e^{ik\lambda} \alpha_k dZ_X(\lambda) \\ &= \int_{-\pi}^{\pi} e^{-it\lambda} \alpha(e^{i\lambda}) dZ_X(\lambda) \end{aligned}$$

Using spectral representation of Y_t , we have

$$Y_t = \int_{-\pi}^{\pi} e^{-it\lambda} dZ_Y(\lambda)$$

Thus, it follows that the random spectral measure of the process Y_t is such that

$$Z_Y(\Lambda) = \int_{\Lambda} \alpha(e^{i\lambda}) dZ_X(\lambda)$$

for all $\Lambda \in \mathcal{B}$. Taking the absolute square and expectation on both sides, we have

$$\begin{aligned} \mathbb{E}|Z_Y(\Lambda)|^2 &= \mathbb{E} \left| \int_{\Lambda} \alpha(e^{i\lambda}) dZ_X(\lambda) \right|^2 \\ &= \mathbb{E} \left| \left(\int_{\Lambda} \alpha(e^{i\lambda}) dZ_X(\lambda) \right) \overline{\left(\int_{\Lambda} \alpha(e^{i\lambda}) dZ_X(\lambda) \right)} \right| \\ &= \int_{\Lambda} \alpha(e^{i\lambda}) \overline{\alpha(e^{i\lambda})} dF_X(\lambda) \\ &= \int_{\Lambda} |\alpha(e^{i\lambda})|^2 dF_X(\lambda) \end{aligned}$$

where again, we have used the isometry of the mapping τ in the third equality. By defining $F_Y(\Lambda) := \mathbb{E}|Z_Y(\Lambda)|^2$, we obtain

$$F_Y(\Lambda) = \int_{\Lambda} |\alpha(e^{i\lambda})|^2 dF_X(\lambda)$$

Assuming absolute continuity of both $F_X(\lambda)$ and $F_Y(\lambda)$ with respect to the Lebesgue measure, we may take Radon-Nykodym derivative w.r.t. the Lebesgue measure to obtain

$$\frac{f_Y(\lambda)}{2\pi} = |\alpha(e^{i\lambda})|^2 \frac{f_X(\lambda)}{2\pi}$$

□

Corollary 1. *Let X_t be an ARMA process given by*

$$\alpha(L)X_t = \beta(L)\varepsilon_t$$

Then, we have

$$|\alpha(e^{i\lambda})|^2 f_X(\lambda) = |\beta(e^{i\lambda})|^2 f_{\varepsilon}(\lambda)$$

1.1 Example

Let ARMA(1,1) process be given as

$$\begin{aligned} X_t &= \alpha X_{t-1} + \varepsilon_t + \beta \varepsilon_{t-1} \\ (1 - \alpha L)X_t &= (1 + \beta L)\varepsilon_t \end{aligned}$$

Using the Theorem above, we have

$$|1 - \alpha e^{i\lambda}|^2 f_X(\lambda) = |1 + \beta e^{i\lambda}|^2 f_\varepsilon(\lambda)$$

Thus, we have

$$\begin{aligned} f_X(\lambda) &= \frac{|1 + \beta e^{i\lambda}|^2}{|1 - \alpha e^{i\lambda}|^2} f_\varepsilon(\lambda) \\ &= \frac{(1 + \beta e^{i\lambda})(1 + \beta e^{-i\lambda})}{(1 - \alpha e^{i\lambda})(1 - \alpha e^{-i\lambda})} \sigma^2 \\ &= \frac{1 + \beta^2 + \beta(e^{i\lambda} + e^{-i\lambda})}{1 + \alpha^2 - \alpha(e^{i\lambda} + e^{-i\lambda})} \sigma^2 \\ &= \frac{1 + \beta^2 + 2\beta \cos(\lambda)}{1 + \alpha^2 - 2\alpha \cos(\lambda)} \sigma^2 \end{aligned}$$

Note that $f_\varepsilon(\lambda) = \sigma^2$ since $f_\varepsilon(\lambda) = \sum_{k=-\infty}^{\infty} \gamma_\varepsilon(k) e^{i\lambda k}$ and $\gamma_\varepsilon(k) = \sigma^2 1\{k = 0\}$.

2 Flipping of the Roots in ARMA processes

For any covariance stationary ARMA processes such that the AR and MA filters do not have roots on the unit circle, we can find a causal and invertible ARMA(p, q) representation with the same second moments. Let X_t be an ARMA(p, q) process given by

$$X_t = \sum_{l=1}^p \alpha_l X_{t-l} + \sum_{l=0}^q \theta_l \varepsilon_{t-l}$$

We may write

$$\alpha(L)X_t = \theta(L)\varepsilon_t$$

where $\varepsilon_t \sim WN(\sigma^2)$.

Assuming that we do not have any roots on the unit circle, both $1/\alpha(z)$ and $1/\theta(z)$ have the Laurent series given in the form

$$\begin{aligned} \frac{1}{\alpha(z)} &= \sum_{n=-\infty}^{\infty} a_n z^n \\ \frac{1}{\theta(z)} &= \sum_{n=-\infty}^{\infty} b_n z^n \end{aligned} \tag{1}$$

which is valid for all z in an open set containing the unit circle. We can further show that we have $\sum_{n=-\infty}^{\infty} |a_n| < \infty$ and $\sum_{n=-\infty}^{\infty} |b_n| < \infty$, which implies that $\alpha(L)^{-1}$ and $\theta(L)^{-1}$

are well-defined since both $\{X_t\}$ and $\{\varepsilon_t\}$ are square integrable. Using the fundamental theorem of algebra, we may factorize $\alpha(L)$ and $\theta(L)$ as

$$\prod_{l=1}^p (I - \alpha_l^{*-1} L) X_t = \prod_{l=1}^q (I - \theta_l^{*-1} L) \varepsilon_t$$

where α_l^* and θ_l^* are roots of the characteristic functions $\alpha(z)$ and $\theta(z)$. The spectral densities of X_t and ε_t are given in the relation

$$\begin{aligned} |\alpha(e^{i\lambda})|^2 f_X(\lambda) &= |\theta(e^{i\lambda})|^2 f_\varepsilon(\lambda) \\ \implies \prod_{l=1}^p |1 - \alpha_l^{*-1} e^{i\lambda}|^2 f_X(\lambda) &= \prod_{l=1}^q |1 - \theta_l^{*-1} e^{i\lambda}|^2 f_\varepsilon(\lambda) \end{aligned}$$

Suppose

$$|\alpha_l^*| = \begin{cases} > 1 & \text{for } l = 1, \dots, r \\ < 1 & \text{for } l = r+1, \dots, p \end{cases}, \quad |\theta_l^*| = \begin{cases} > 1 & \text{for } l = 1, \dots, s \\ < 1 & \text{for } l = s+1, \dots, q \end{cases}$$

Now define

$$\begin{aligned} \tilde{\alpha}(L) &= \prod_{1 \leq l \leq r} (I - a_l^{*-1} L) \prod_{r < l \leq p} (I - \overline{a_l^*} L) \\ \tilde{\theta}(L) &= \prod_{1 \leq l \leq s} (I - \theta_l^{*-1} L) \prod_{s < l \leq q} (I - \overline{\theta_l^*} L) \end{aligned}$$

Consider the ARMA(p, q) process $\{Y_t\}$ defined by

$$\tilde{\alpha}(L)Y_t = \tilde{\theta}(L)\varepsilon_t$$

$\{Y_t\}$ is both causal and invertible by construction and has the spectral density

$$\prod_{l=1}^r |1 - \alpha_l^{*-1} e^{i\lambda}|^2 \prod_{l=r+1}^p |1 - \overline{\alpha_l^*} e^{i\lambda}|^2 f_Y(\lambda) = \prod_{l=1}^s |1 - \theta_l^{*-1} e^{i\lambda}|^2 \prod_{l=s+1}^q |1 - \overline{\theta_l^*} e^{i\lambda}|^2 f_\varepsilon(\lambda)$$

Since

$$|1 - \overline{a_l^*} e^{i\lambda}| = |1 - a_l^* e^{-i\lambda}| = |a_l^*| |e^{-i\lambda}| |1 - a_l^{*-1} e^{i\lambda}| = |a_l^*| |1 - a_l^{*-1} e^{i\lambda}|$$

we can rewrite the spectral density as

$$\prod_{l=1}^p |1 - \alpha_l^{*-1} e^{i\lambda}|^2 \left(\prod_{l=r+1}^p |a_l^*|^2 \right) f_Y(\lambda) = \prod_{l=1}^q |1 - \theta_l^{*-1} e^{i\lambda}|^2 \left(\prod_{l=s+1}^q |\theta_l^*|^2 \right) f_\varepsilon(\lambda)$$

Thus, we have

$$\begin{aligned} f_Y(\lambda) &= \frac{\prod_{l=s+1}^q |\theta_l^*|^2}{\prod_{l=r+1}^p |a_l^*|^2} \left(\frac{\prod_{l=1}^q |(1 - \theta_l^{*-1} e^{i\lambda})|^2}{\prod_{l=1}^p |(1 - \alpha_l^{*-1} e^{i\lambda})|^2} f_\varepsilon(\lambda) \right) \\ &= \frac{\prod_{l=s+1}^q |\theta_l^*|^2}{\prod_{l=r+1}^p |a_l^*|^2} f_X(\lambda) \end{aligned}$$

We see that the new ARMA(p, q) process given by

$$\tilde{\alpha}(L)\tilde{X}_t = \tilde{\theta}(L)\tilde{\varepsilon}_t$$

where $\tilde{\varepsilon}_t$ is white noise with variance $\frac{\prod_{l=r+1}^p |a_l^*|^2}{\prod_{l=s+1}^q |\theta_l^*|^2} \sigma^2$ has exactly the same spectral density as $f_X(\lambda)$ and also is both causal and invertible.

2.1 Examples

1. Consider a MA(1) process given by

$$X_t = \varepsilon_t + \theta\varepsilon_{t-1}$$

We have

$$\gamma(k) = \begin{cases} (1 + \theta^2)\sigma^2, & k = 0 \\ \theta\sigma^2, & |k| = 1 \\ 0, & |k| > 1 \end{cases}$$

How do we obtain a process whose root has been flipped? Notice that

$$\theta(z) = 1 + \theta z$$

has root given by $z^* = -\frac{1}{\theta}$. Consider the process obtained after we flip the root.

$$Y_t = \varepsilon_t + \frac{1}{\theta}\varepsilon_{t-1}$$

Now, adjust for the variance change due to the flipping of the root by using a new white noise process $\tilde{\varepsilon}_t = |\theta|\varepsilon_t$. Then, we have

$$\begin{aligned} \tilde{X}_t &= \tilde{\varepsilon}_t + \frac{1}{\theta}\tilde{\varepsilon}_{t-1} \\ &= |\theta|\varepsilon_t + \frac{|\theta|}{\theta}\varepsilon_{t-1} \end{aligned}$$

This process has exactly the same second moments as the original process (X_t). Depending on $|\theta| > 1$ or $|\theta| < 1$, one is invertible while the other is not.

2. Consider a stationary AR(1) process given by

$$X_t = \alpha X_{t-1} + \varepsilon_t$$

whose second moments are

$$\gamma(k) = \begin{cases} \sigma^2 \frac{\alpha^{|k|}}{1-\alpha^2}, & |\alpha| < 1 \\ \sigma^2 \frac{1}{\alpha^{|k|}(\alpha^2-1)}, & |\alpha| > 1 \end{cases}$$

To get the other representation, consider

$$\alpha(z) = 1 - \alpha z$$

which has root given by $z^* = \frac{1}{\alpha}$. Consider a process with the root flipped.

$$Y_t = \frac{1}{\alpha} Y_{t-1} + \varepsilon_t$$

Now adjust for the variance by using $\tilde{\varepsilon}_t = \frac{1}{|\alpha|} \varepsilon_t$.

$$\begin{aligned} \tilde{X}_t &= \frac{1}{\alpha} \tilde{X}_{t-1} + \tilde{\varepsilon}_t \\ &= \frac{1}{\alpha} \tilde{X}_{t-1} + \frac{1}{|\alpha|} \varepsilon_t \end{aligned}$$

Note that this process has exactly the same second moments as the original X_t . Depending on $|\alpha| > 1$ or $|\alpha| < 1$, one is invertible while the other is not.

3 Ergodic Theorem

Let (X_t) be a mean zero stationary process and let $\mathcal{M} = \text{span}(\{X_t\}_{t=-\infty}^{\infty})$. Let $\mathcal{M}_0$ be a subspace of $\mathcal{M}$ that is time invariant, i.e.,

$$\mathcal{M}_0 = \{X \in \mathcal{M} : LX = X\}$$

Then, we have

$$\frac{1}{n} \sum_{t=1}^n X_t \rightarrow_{L^2} P_0 X_0$$

where P_0 is projection on $\mathcal{M}$.

Proof. Let $\mathcal{N}(T)$ and $\mathcal{R}(T)$ be the null space and the range of an operator $T : \mathcal{M} \rightarrow \mathcal{M}$.

$$\begin{aligned} \mathcal{N}(T) &= \{X \in \mathcal{M} : TX = 0\} \\ \mathcal{R}(T) &= \{Y \in \mathcal{M} : Y = TX, X \in \mathcal{M}\} \end{aligned}$$

We see that $\mathcal{M}_0$ is a null space of the operator $L - I$. We may further show that it is the orthogonal complement to the range of the operator $L^{-1} - I$. In other words, we have

$$\mathcal{M}_0 = \mathcal{N}(L - I) = \mathcal{R}(L^{-1} - I)^\perp$$

where

$$\mathcal{R}(L^{-1} - I)^\perp = \{X \in \mathcal{M} : \langle X, Y \rangle = 0, \forall Y \in \mathcal{R}(L^{-1} - I)\}$$

To show this, we show that the set of X that satisfies

$$\langle X, (L^{-1} - I)Y \rangle = 0$$

for all $Y \in \mathcal{M}$ is the null space $\mathcal{N}(L - I)$. Note that we have

$$\langle X, (L^{-1} - I)Y \rangle = \langle (L - I)X, Y \rangle \tag{2}$$

since both L and I are unitary operators. We may also show (2) directly.

$$\begin{aligned}
\langle X_t, (L^{-1} - I)Y_t \rangle &= \mathbb{E}[X_t(Y_{t+1} - Y_t)] \\
&= \mathbb{E}[X_t Y_{t+1}] - \mathbb{E}[X_t Y_t] \\
&= \mathbb{E}[X_{t-1} Y_t] - \mathbb{E}[X_t Y_t] \\
&= \mathbb{E}[(X_{t-1} - X_t)Y_t] \\
&= \langle (L - I)X_t, Y_t \rangle
\end{aligned}$$

Now, we see that the RHS of (2) is zero for all $Y \in \mathcal{M}$ if and only if $X \in \mathcal{N}(L - I)$. Thus we have shown $\mathcal{N}(L - I) = \mathcal{R}(L^{-1} - I)^\perp$. Using the projection theorem, we have

$$\begin{aligned}
X_0 &= P_0 X_0 + (I - P_0)Y \\
&= P_0 X_0 + (L^{-1} - I)Y
\end{aligned}$$

for some $Y \in \mathcal{M}$ where P_0 is projection on $\mathcal{M}$. Note that

$$\begin{aligned}
\frac{1}{n} \sum_{t=1}^n X_t &= \left(\frac{1}{n} \sum_{t=1}^n L^{-t} \right) X_0 \\
&= P_0 X_0 + \left(\frac{1}{n} \sum_{t=1}^n L^{-t} \right) (L^{-1} - I)Y \\
&= P_0 X_0 + \frac{1}{n} (L^{-(n+1)} - L^{-1}) Y
\end{aligned}$$

By triangle inequality, we have

$$\left\| \frac{1}{n} \sum_{t=1}^n X_t \right\| \leq P_0 X_0 + \left\| \frac{1}{n} (L^{-(n+1)} - L^{-1}) Y \right\|$$

To complete the proof, notice that we have

$$\begin{aligned}
\left\| \frac{1}{n} (L^{-(n+1)} - L^{-1}) Y \right\|^2 &= \frac{1}{n^2} \|L^{-(n+1)}Y - L^{-1}Y\|^2 \\
&\leq \frac{4}{n^2} \|Y\|^2 \rightarrow 0
\end{aligned}$$

due to Cauchy-Swartz inequality. In general, we may show that $(\sum_{i=1}^n a_i)^2 \leq n(\sum_{i=1}^n a_i^2)$ using Cauchy-Swartz inequality on $(a_1, \dots, a_n)$ and $(1, \dots, 1)$. $\square$

4 Estimation of spectral density function

Let $Z_n(\lambda) = \frac{1}{n} \sum_{t=1}^n X_t e^{i\lambda t}$ be finite Fourier transform of X_t . Periodogram $I_n(\lambda)$ is defined as

$$I_n(\lambda) = n|Z_n(\lambda)|^2 = \frac{1}{n} \left| \sum_{t=1}^n X_t e^{it\lambda} \right|^2$$

We have

$$\begin{aligned}
\frac{1}{n} \left| \sum_{t=1}^n X_t e^{it\lambda} \right|^2 &= \frac{1}{n} \left(\sum_{t=1}^n X_t e^{it\lambda} \right) \left(\sum_{t=1}^n X_t e^{-it\lambda} \right) \\
&= \hat{\gamma}(-n+1)e^{-i(n-1)\lambda} + \hat{\gamma}(-n+2)e^{-i(n-2)\lambda} + \dots + \hat{\gamma}(0)e^0 + \dots + \hat{\gamma}(n-1)e^{i(n-1)\lambda} \\
&= \sum_{|k| < n} \hat{\gamma}(k) e^{ikn}
\end{aligned}$$

where $\hat{\gamma}(k) = \frac{1}{n} \sum_{t=k+1}^n X_t X'_{t-k}$. Thus, a natural estimator for $f(\lambda) = \sum_{k=-\infty}^{\infty} \gamma(k) e^{ik\lambda}$ is $I_n(\lambda)$. However, we do not have $I_n(\lambda) \rightarrow_p f(\lambda)$. This is because the speed at which the number of estimates $\{\hat{\gamma}(k)\}_{|k| < n}$ increases at the same rate as the speed at which sample size n increases. For any n , there will exist $\hat{\gamma}(n-1)$, which consists of only one sample. Thus, we use

$$\hat{f}_n(\lambda) = \sum_{|k| < l_n} w\left(\frac{k}{l_n}\right) \hat{\gamma}(k) e^{ik\lambda}$$

where $w(\cdot)$ is called a kernel and l_n a lag truncation number. We chose l_n so that $\frac{l_n}{n} \rightarrow 0$ as $n \rightarrow \infty$ so that $\{\hat{\gamma}(k)\}_{|k| < l_n}$ will be consistent. However, we also let $l_n \rightarrow \infty$ as $n \rightarrow \infty$ so that $\hat{f}_n(\lambda) \rightarrow f(\lambda)$. The weights $w(\cdot)$ act so that we give less weights to $\hat{\gamma}(k)$ for k large because we know these estimates are unstable.

In estimating spectral density functions, we need to choose both the lag truncation number l_n and the kernel $w(\cdot)$. Which kernel to use does not have a large effect on the final estimate $f_n(\lambda)$.

1. Having large l_n reduces bias but increases variance due to large variances in the estimates $\hat{\gamma}(k)$ for k large.
2. Having small l_n reduces variance but introduces bias from truncation.